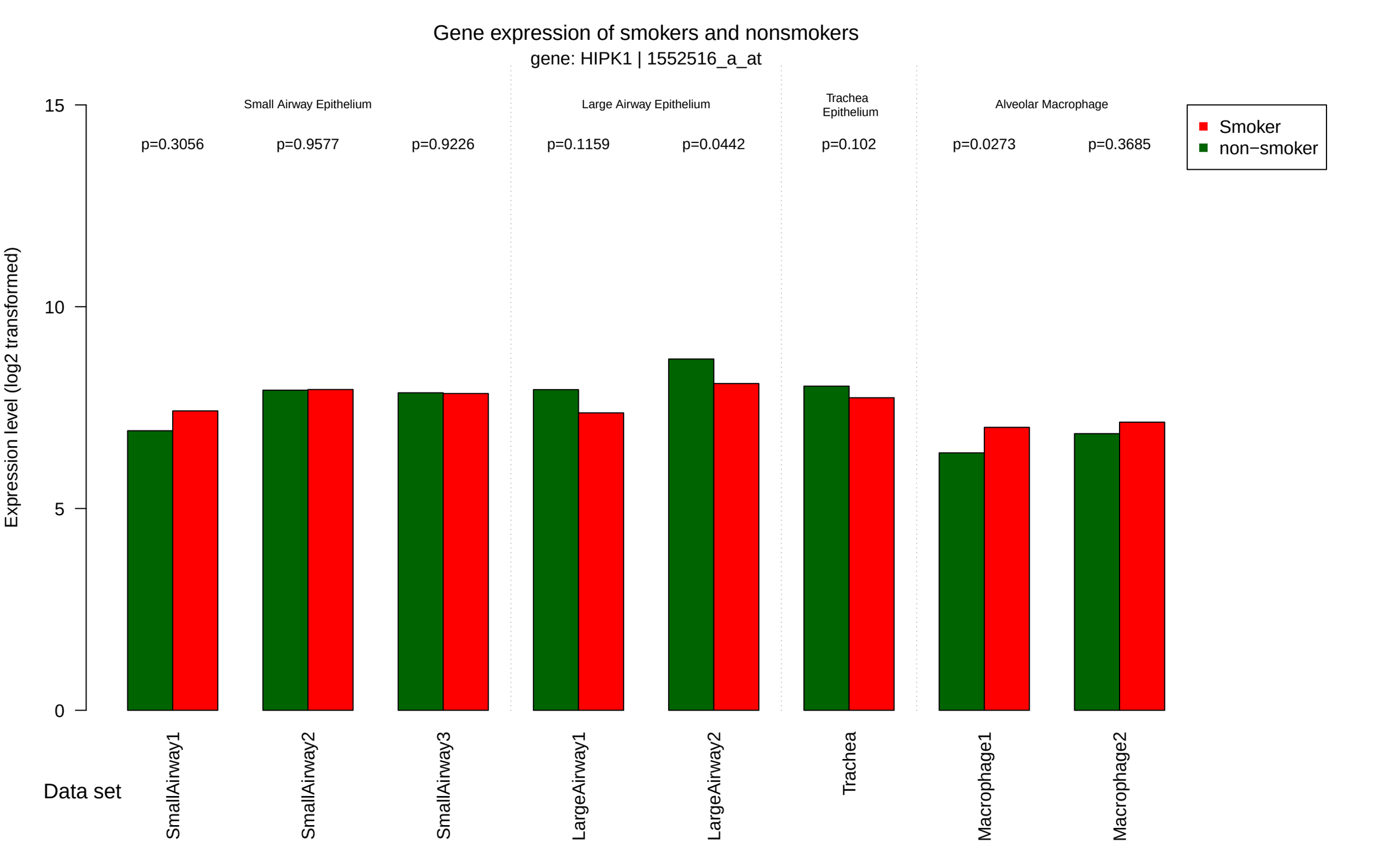
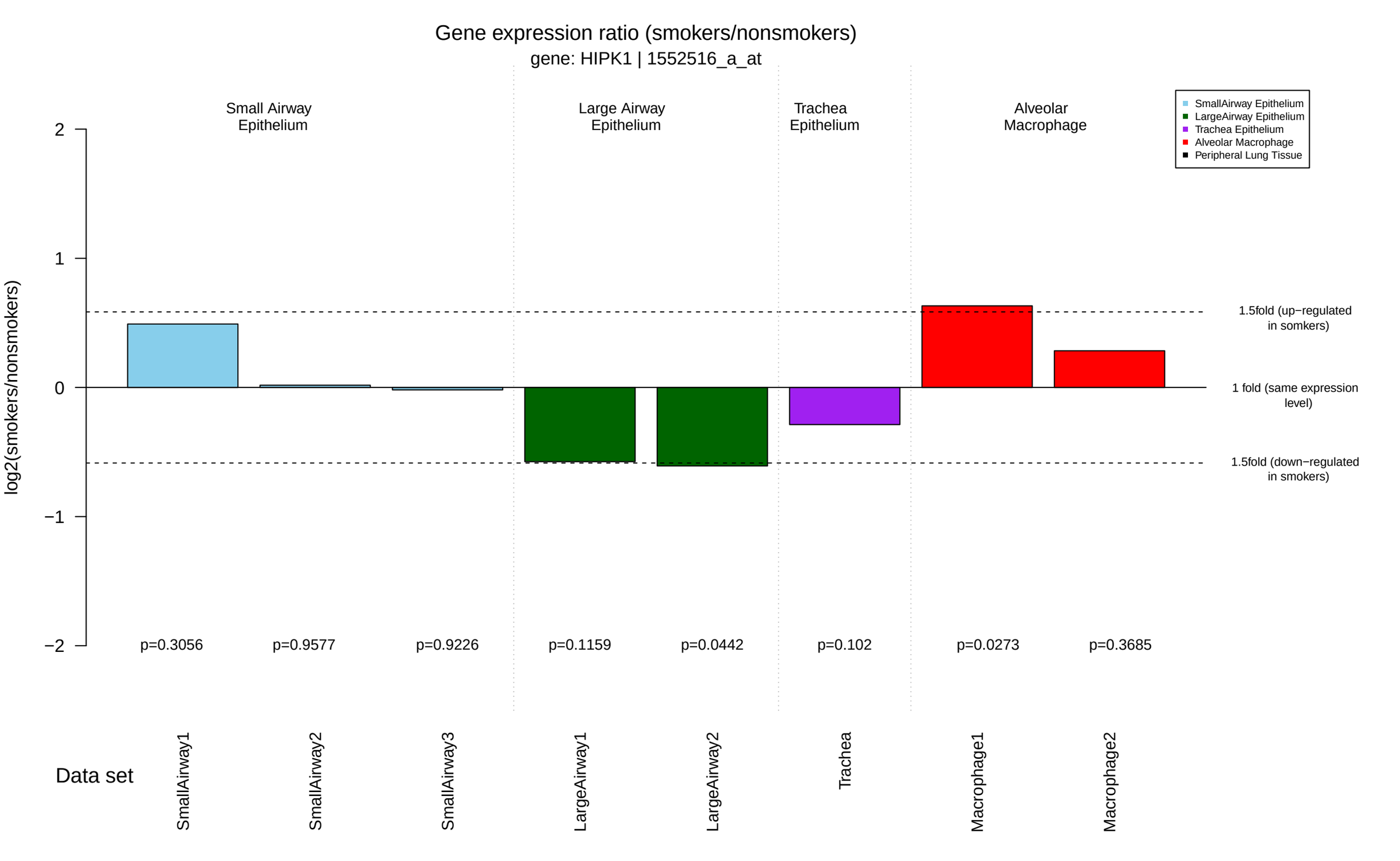
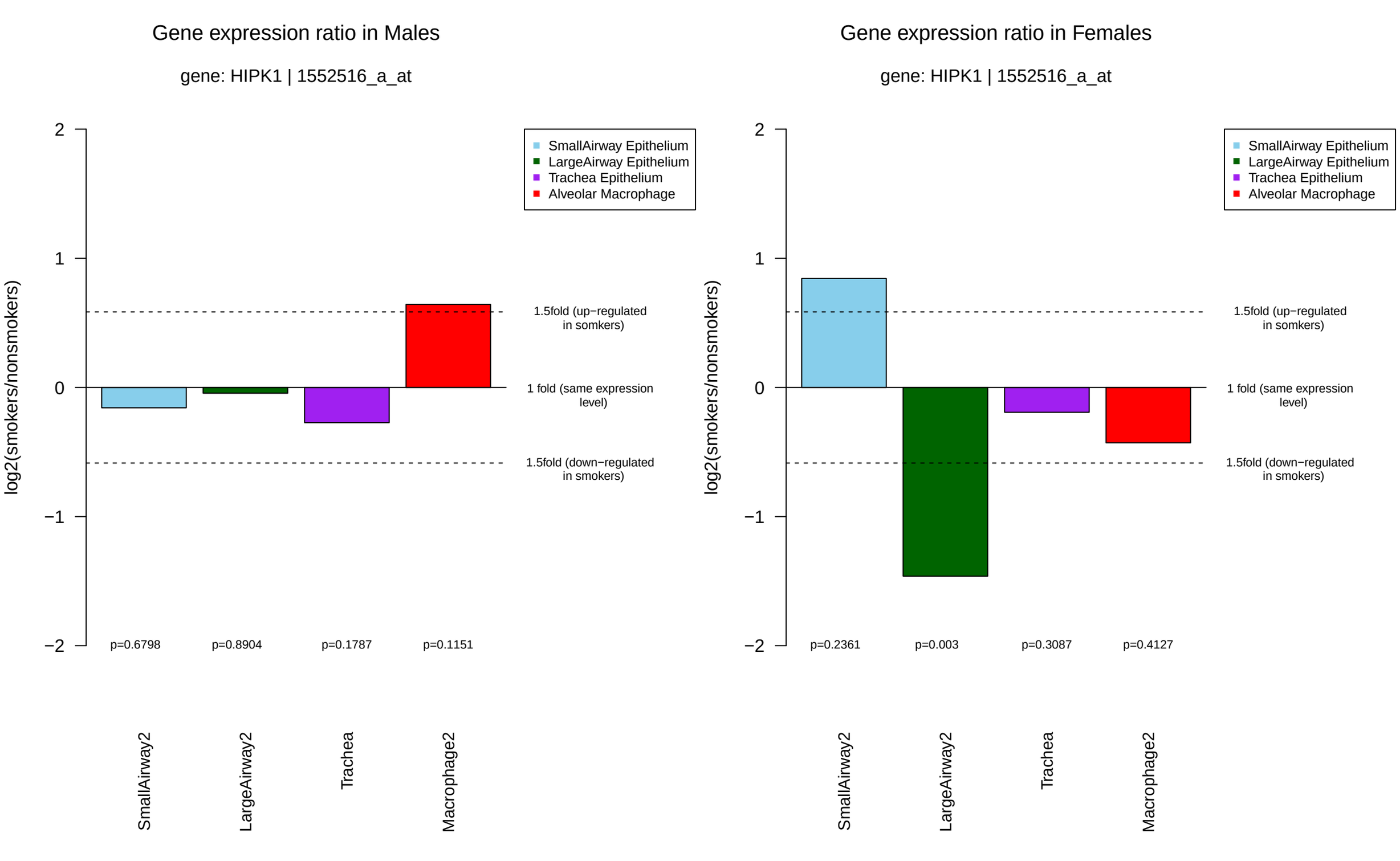
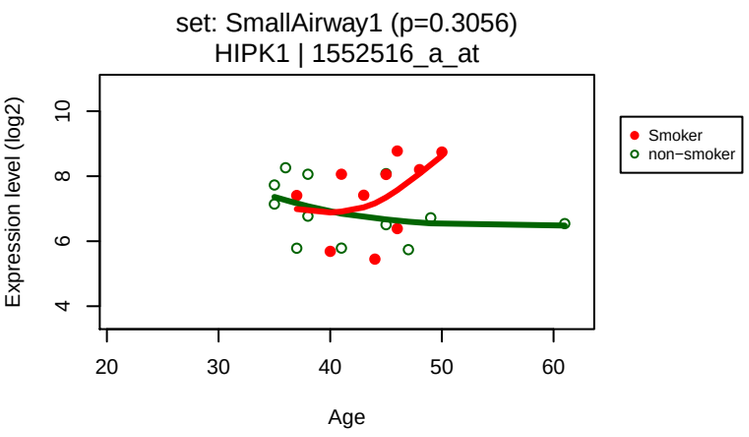
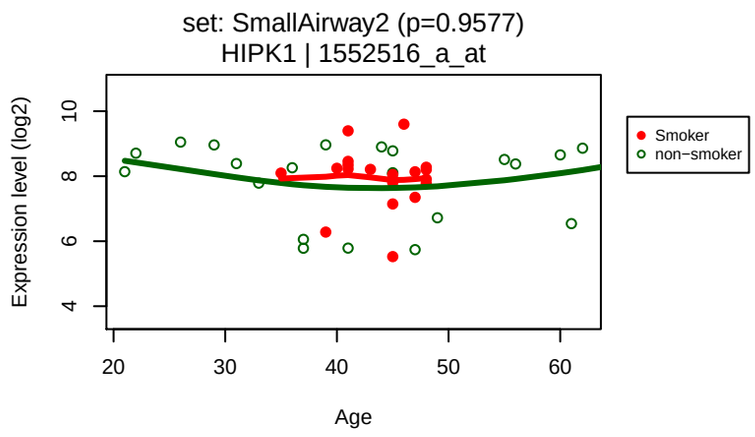
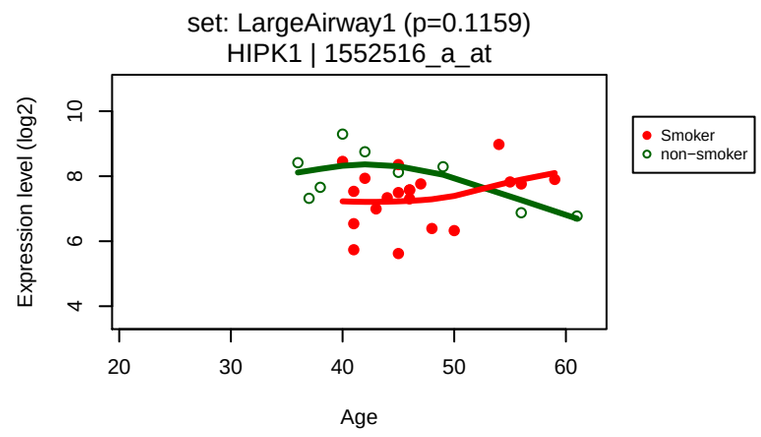

Gene expression of smokers and nonsmokers
gene: HIPK1 | 1552516_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3056
p=0.9577
p=0.9226
p=0.1159
p=0.0442
p=0.102
p=0.0273
p=0.3685
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

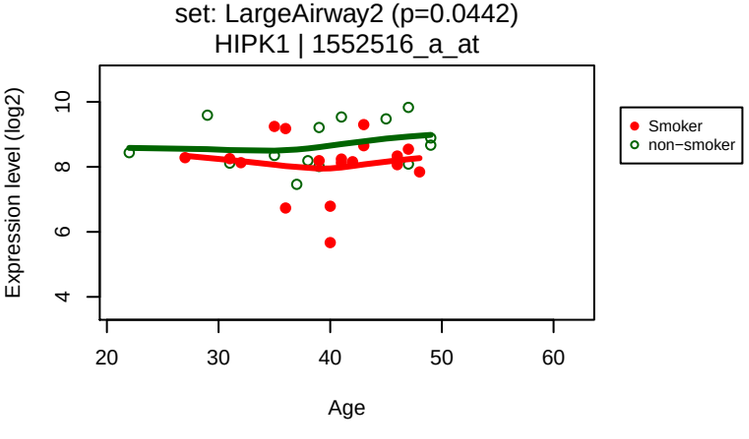

Gene expression ratio (smokers/nonsmokers)
gene: HIPK1 | 1552516_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3056
p=0.9577
p=0.9226
p=0.1159
p=0.0442
p=0.102
p=0.0273
p=0.3685
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

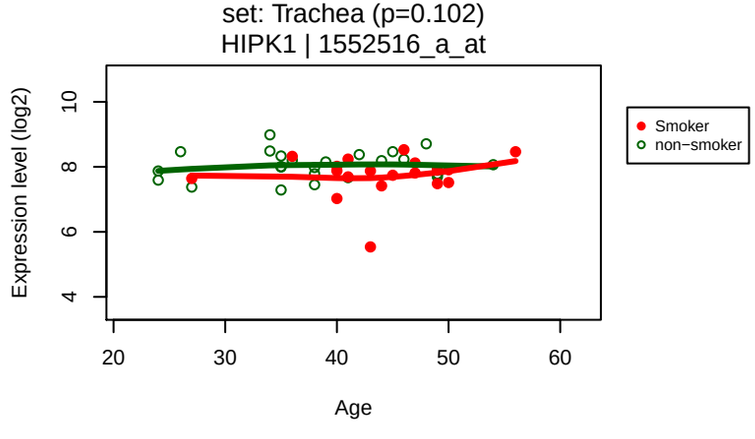

Gene expression ratio in Males
gene: HIPK1 | 1552516_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6798
p=0.8904
p=0.1787
p=0.1151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK1 | 1552516_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2361
p=0.003
p=0.3087
p=0.4127
SmallAirway2
LargeAirway2
Trachea
Macrophage2

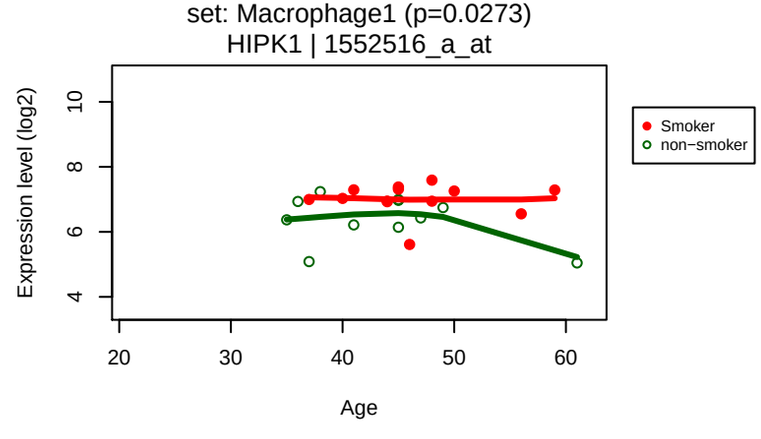

set: SmallAirway1 (p=0.3056)
HIPK1 | 1552516_a_at
Expression level (log2)
Age
Smoker
non−smoker

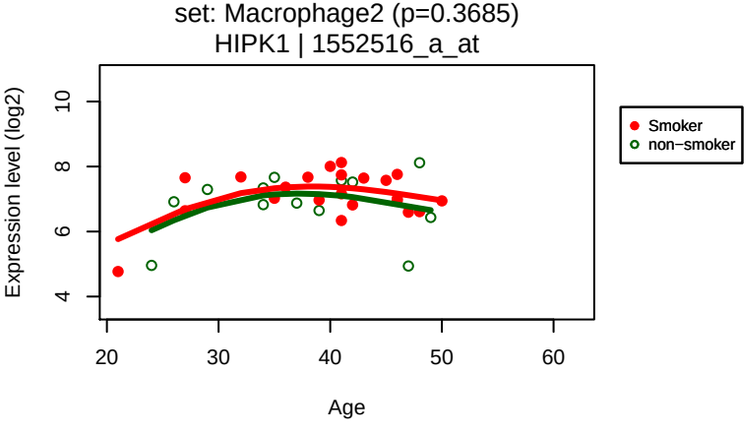

set: SmallAirway2 (p=0.9577)
HIPK1 | 1552516_a_at
Expression level (log2)
Age
Smoker
non−smoker

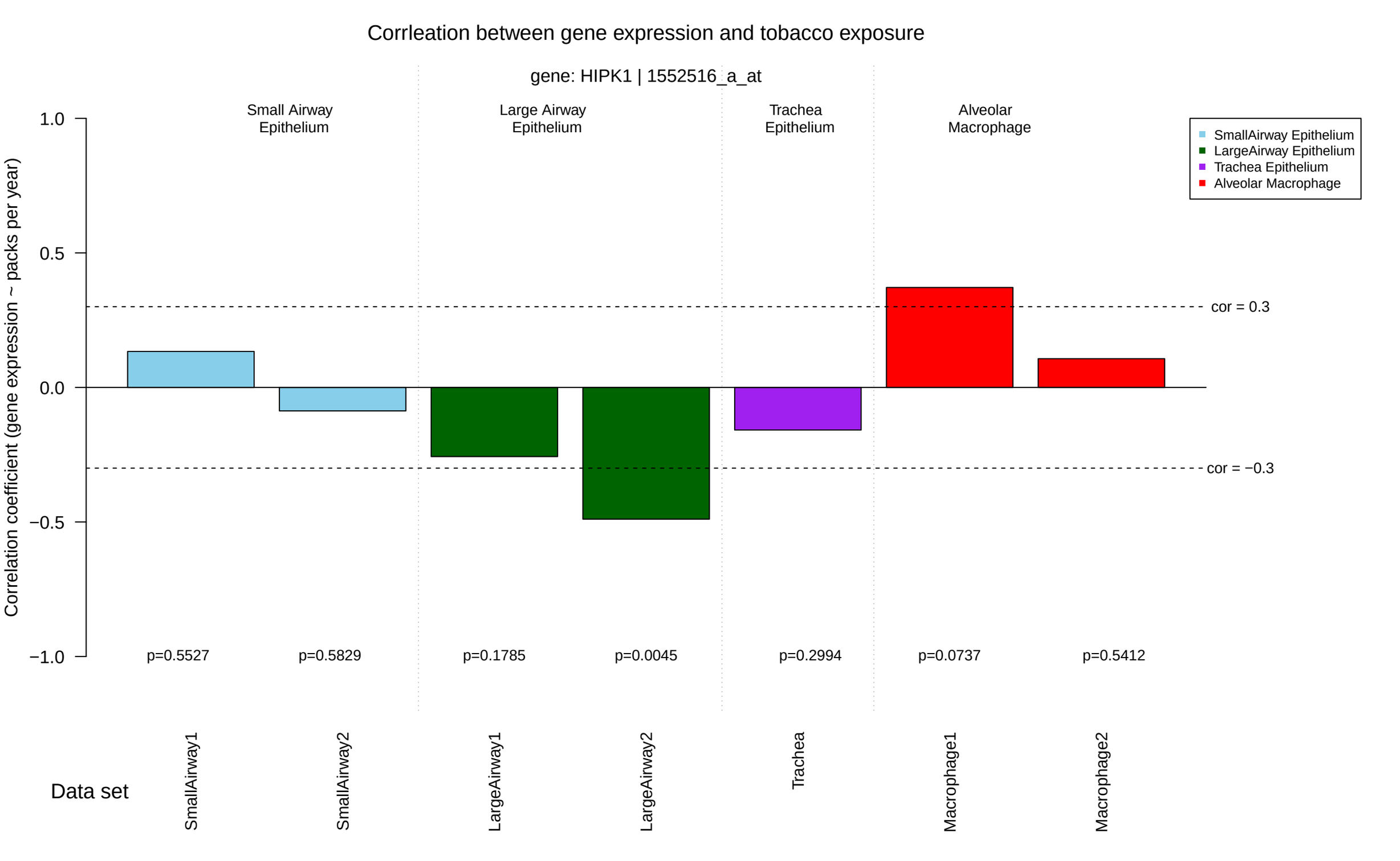

set: LargeAirway1 (p=0.1159)
HIPK1 | 1552516_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0442)
HIPK1 | 1552516_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.102)
HIPK1 | 1552516_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0273)
HIPK1 | 1552516_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3685)
HIPK1 | 1552516_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK1 | 1552516_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5527
p=0.5829
p=0.1785
p=0.0045
p=0.2994
p=0.0737
p=0.5412
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2